

5(a)	joule per coulomb	B1
5(b)(i)	$1/R = 1/R_1 + 1/R_2$ $= 1/300 + 1/200$ $R = 75 \Omega$	A1
5(b)(ii)	$R = 75 + 55$ $= 130 \Omega$	A1
5(c)(i)	1. $P = I^2 R$ or $P = VI$ and $V = IR$	C1
	$I = (0.20/55)^{0.5}$ $= 0.060 \text{ A}$	A1
	2. $I = 0.060/4$ $= 0.015 \text{ A}$	A1
5(c)(ii)	potential difference $= 130 \times 0.060$ $= 7.8 \text{ V}$	A1
	or	
	potential difference $= (300 \times 0.015) + (55 \times 0.060)$ $= 7.8 \text{ V}$ (<i>other valid methods are also possible</i>)	(A1)